

Basics of storage - NAS, DAS and SAN-03

1. Introduction to Storage

STORAGE TYPES COMPARISON



Storage systems are used to save, organize, and manage data for personal, business, and enterprise needs. Choosing the right storage type depends on performance, scalability, cost, and access requirements. The three main types are **Direct Attached Storage (DAS)**, **Network Attached Storage (NAS)**, and **Storage Area Network (SAN)**.

2. Direct Attached Storage (DAS)

- **Definition:** Storage directly connected to a single computer or server through interfaces like SATA, SCSI, or NVMe.
- **Examples:** Internal hard drives, SSDs, or external USB drives.
- **Advantages:**
 - Simple and low-cost.
 - Easy to install and use.
 - Good performance for single systems.
- **Limitations:**
 - Limited scalability.
 - Data is not easily shared across multiple systems.
 - Management becomes difficult as data grows.
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3. Network Attached Storage (NAS)

- **Definition:** A dedicated storage device connected to a network, allowing multiple users and devices to access files. Works at the **file level** using protocols like NFS, SMB, or FTP.
- **Examples:** Synology NAS, QNAP NAS devices.
- **Advantages:**
 - Centralized storage accessible over LAN/WAN.
 - Easy file sharing and collaboration.
 - Supports backup and media streaming.
 - Scalable with additional drives or enclosures.
- **Limitations:**
 - Performance depends on network speed.
 - Not suitable for high-performance transactional workloads.

4. Storage Area Network (SAN)

- **Definition:** A high-speed network of storage devices providing block-level access to servers. Uses protocols like Fibre Channel (FC) or iSCSI.
- **Examples:** Enterprise SANs used in data centers.
- **Advantages:**
 - High performance and low latency.
 - Centralized and scalable storage pool.
 - Suitable for databases, virtualization, and mission-critical applications.
- **Limitations:**
 - Expensive and complex to set up.
 - Requires specialized hardware and administration.